

ASTRODYNAMICS I: HOMEWORK #1

Due: Mon, Mar 26, 11:00AM, 2018

1. A satellite is in a circular parking orbit about the Earth (120km altitude). To begin a voyage to the Moon, what velocity does it need to escape from the Earth?

Would your answer be different for an elliptical orbit ($e = 0.3$, $\nu = 145^\circ$, a the same)?

Why or why not?

If you assume both burns are tangent to the existing velocity, what are the required increases in velocity for each case?

2. Is there a common point at which the velocity of a satellite in elliptical orbit equals the velocity in a circular orbit? If so, where?

3. Identify the type of orbits given the following data:

a. $\vec{r} = 6872.0\hat{i} + 6872.0\hat{j} \text{ km}$, $\vec{v} = -4.7028\hat{i} + 4.7028\hat{j} \text{ km/sec}$

b. $\vec{e} = 0.08\hat{i} + 0.45\hat{j} + 0.021\hat{k}$

c. $\vec{h} = 1.6\hat{j} \text{ ER}^2/\text{TU}$

d. $\vec{h} = -3.5\hat{k} \text{ ER}^2/\text{TU}$

4. Holmes comet moves in an elliptical orbit of eccentricity 0.433 and perihelion distance 2.053 AU. Compare its velocities, both linear and angular, at perihelion and aphelion.

5. Consider a satellite in a “nearly circular” orbit about the Earth.
 - a. Plot the gravitational acceleration of the satellite due to Earth’s gravity as a function of the altitude of the satellite as measured from the surface of the Earth. Scale your plot so that the altitude goes from 0 km to 10,000 km.
 - b. Assume a satellite lies on the line between the Earth and the Sun. Make two plots of the gravitational acceleration of the satellite due to Earth’s gravity and of the gravitational acceleration of the satellite due to the Sun’s gravity, each as a function of the altitude of the satellite as measured from the surface of the Earth. Scale your plot so that the satellite altitude goes from 0 km to 10 million km.
 - c. Using your plots in part (b), assess the relative importance of Earth’s gravity and Sun’s gravity on the satellite’s orbit. For what satellite altitudes do you think it is a reasonable approximation to ignore the Sun’s gravity and use a two-body model for the satellite’s orbit based only on Earth’s gravity?

6. At a specific instant of time, an orbiting spacecraft is at an altitude of 1500 km above the Earth’s surface; at this instant the velocity vector of the spacecraft has magnitude 7.8 km/ sec and flight path angle of 4.5 degrees.
 - a. What are the values of the local vertical velocity component $\dot{r}$ and the local horizontal velocity component $r\dot{\theta}$ at this instant?
 - b. What is the specific energy of this orbit?
 - c. What is the specific angular momentum of this orbit?
 - d. What type of conic section is the orbit?

7. The Energy and Angular Momentum of several asteroids heading towards Earth have been measured. Using the Energy and Angular Momentum integrals only, determine which of these asteroids are a real problem, i.e., which ones will hit the Earth?

Hint: The angular momentum when closest to the Earth is $h = r_p v_p$, where r_p is the closest radius to the Earth, and v_p is the speed when closest to the Earth.

Asteroid #	Energy (km^2/s^2)	Angular Momentum (km^2/s)
1	1	1×10^5
2	100	1×10^5
3	0	7×10^4
4	0	8×10^4
5	10	8×10^4